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A cautionary note on the use of the Hypermet tailing function in X-ray spectrometry with Si(Li) detectors

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Abstract

The widely used Hypermet function that describes quasi-exponential tailing in Si(Li) X-ray detectors usually has a multiplicative factor omitted. While this omission does not affect the quality of least-squares fits to spectra, it can cause error in the deduced intensity ratio between the tail and its parent Gaussian.

1. Introduction

The spectral response of Si(Li) X-ray spectrometers is predominantly Gaussian, with a low-energy component whose relative intensity depends in part upon the detector fabrication process. In elemental analysis techniques based upon X-ray emission spectrometry, this response function must be accurately modelled for the purpose of least-squares fitting the measured spectra, and thereby extracting the various peak intensities [1].

Many authors refer to the important early work of Phillips and Marlow [2], in which the so-called HYPERMET function was applied successfully to describe the low-energy components. For an ideal detector, a photon with negligible natural width would produce a sharp line (delta function) in the spectrum; in a real detector this line is broadened by electronic noise and the statistics of the ionization process into a Gaussian shape, as shown in the upper panel of Fig. 1. Phillips and Marlow [2] suggested that the effect of incomplete charge collection would remove counts from the main peak, “producing in an ideal detector an exponentially decaying distribution below the peak”. But this feature again must be broadened by the effects of noise and statistics, as in the second panel of Fig. 1. A third feature, invariably observed, is an essentially flat shelf running from the main peak down to very low energy; this is shown in the third panel. Experiments by various authors using monoenergetic photons (see Ref. [3]) show that the sum of these three broadened features (right side of Fig. 1) provides an excellent description of the measured lineshape.

2. Detailed discussion

We describe the Gaussian $G(i)$, the simple exponential $E(i)$ and the convoluted exponential $D(i)$ as follows, with i representing the channel number and i_0 the centroid channel of the parent Gaussian. These are the expressions given in Ref. [2], but converted to a nomenclature that is now rather widely used.

$$G(i) = H_G \exp \left[\frac{-(i - i_0)^2}{2\sigma^2} \right], \quad (1)$$

$$E(i) = H_D \exp \left(\frac{i - i_0}{\beta} \right), \quad i < i_0, \\ = 0, \quad i \geq i_0, \quad (2)$$

$$D(i) = \frac{1}{2} H_D \exp \left(\frac{i - i_0}{\beta} \right) \operatorname{erfc} \left(\frac{i - i_0}{\sigma\sqrt{2}} + \frac{\sigma}{\beta\sqrt{2}} \right). \quad (3)$$

Here H_G is the height of the Gaussian, σ is the standard deviation of the Gaussian, H_D is the height of the exponential at the Gaussian centroid, and β is the slope of the exponential. It is customary to use the ratios $R_H = H_D/H_G$ and $R_S = \beta/\sigma$ as the descriptors of the tail in any fitting procedure, because these two quantities vary smoothly with X-ray energy, reflecting the properties of the particular Si(Li) detector.

In describing the detector's relative efficiency, one requires the intensity or area ratio of the features D and G ; from Eqs. (1) and (2) this is simply

$$\frac{A_D}{A_G} = \frac{1}{\sqrt{2\pi}} R_S R_H. \quad (4)$$

Because the convolution in Eq. (3) is effected using a unit-area Gaussian, the result of Eq. (4) is not altered by the convolution.

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We recently observed in practice that when $R_S < 2$, i.e. for narrow tails of only modest asymmetry, Eq. (4) gives results that appear to be anomalous. To test this observation, we generated tailing functions by the simple expedient of numerically convoluting a pure exponential with a unit-area Gaussian function. We then computed the area of this tailing feature first by using Eq. (4) and second by summing the contents of the occupied channels in the simulated spectrum. The lower, full curve in Fig. 2 shows the ratio of the tail areas obtained by these two approaches. This ratio should be unity, but in fact falls rapidly below unity for R_S values less than 3.

We therefore re-calculated the convolution integral (see Appendix), and found

$$D(i) = \frac{1}{2} H_D \exp\left(\frac{i - i_0}{\beta}\right) \exp\left(\frac{\sigma^2}{2\beta^2}\right) \times \operatorname{erfc}\left(\frac{i - i_0}{\sigma\sqrt{2}} + \frac{\sigma}{\beta\sqrt{2}}\right). \quad (5)$$

This expression differs from Eq. (3) by the factor $\exp(\sigma^2/2\beta^2)$. This factor corresponds closely to the error factor revealed in Fig. 2.

Use of the older form (3) of $D(i)$ has no deleterious effects upon the quality of fit, as compared to the new version (5). The version given originally by Phillips and Marlow is a legitimate one. But it now becomes obvious that by excluding the term $\exp(\sigma^2/2\beta^2)$, or, to be more precise, by incorporating it within their tail height parameter, these authors rendered their tail height parameter a function of the tail slope. This has two disadvantageous outcomes.

The first outcome is the introduction of a correlation between the fitting parameters that describe the exponen-

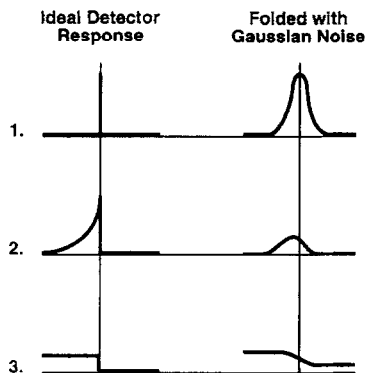


Fig. 1. Idealized (left) and resolution-broadened (right) peak components. Upper panel: the delta function is the main response component of a perfect Si(Li) detector to monoenergetic X-rays, and the Gaussian is the convolute of that response with the Gaussian resolution function. Middle panel: exponential tail results from detector imperfections. Lower panel: flat shelf results from electron escape.

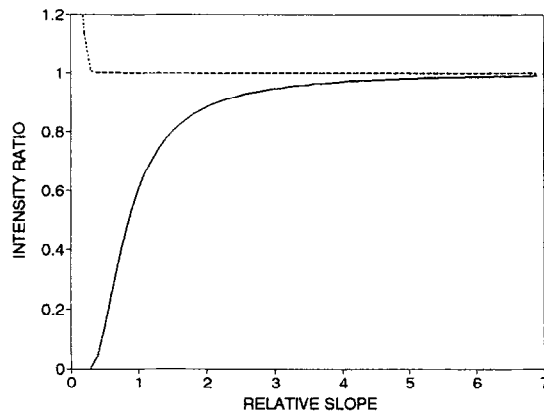


Fig. 2. Ratio of summed tail intensity to that computed by Eq. (4); lower, full curve is obtained when the tail is represented by Eq. (3); upper, dashed curve is the result of using Eq. (5).

tial tail, viz. R_S and R_H . In addition, the partial derivatives of the fit function with respect to the slope parameter will not be correct, because a term involving the slope has been omitted from the function. Explicit inclusion in the overall model function of the factor $\exp(\sigma^2/2\beta^2)$ should make the fitting more robust with respect to the tail function.

The second outcome of omitting the factor $\exp(\sigma^2/2\beta^2)$ from the convoluted tail function is that Eq. (4) generates an erroneous value of the ratio of exponential tail intensity to Gaussian intensity. (The intensity ratio of Eq. (4) must be multiplied by $\exp(-\sigma^2/2\beta^2)$ to give the correct result.) The large error in relative tail intensity when $R_S < 2$ will introduce a corresponding error into the relative efficiency of the detector, as this quantity must include a factor describing those peak counts that appear in the tail instead of the Gaussian.

We find that when the tail is properly described in our GUPIX code [4,5] for fitting PIXE spectra, that the tail intensity is now calculated correctly except at very low values of R_S below 0.3. This is shown as the upper, dashed curve in Fig. 2. The residual error at low R_S values is due to the inherent accuracy limit of about $3E-7$ in our numerical method of computing the complementary error function. We have therefore restricted the parameter R_S to values in excess of 0.3 in GUPIX.

3. Conclusions

The widely used HYPERMET function has to date lacked a multiplicative factor. This neglect has no effect on the shape of the Gaussian-convoluted exponential tail, but it causes error in the computed intensity of the tail relative to that of its parent Gaussian. This in turn causes error in the deduced intrinsic efficiency of the Si(Li) detector. Inclusion of the omitted factor rectifies these errors and

improves the robustness of the parameters generated by the least-squares fit.

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Appendix A

To simplify the derivation of Eq. (5), we set $x = i - i_0$ and consider a Gaussian of unit area. Thus:

$$G(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(\frac{-x^2}{2\sigma^2}\right).$$

The exponential tail is then

$$E(x) = \begin{cases} H_D \exp(x/\beta) & \text{for } x \leq 0 \\ 0 & \text{for } x > 0 \end{cases}.$$

The convolution is

$$\begin{aligned} D(x) &= \int_{-\infty}^{\infty} G(x-t)E(t)dt \\ &= \int_{-\infty}^0 G(x-t)E(t)dt \end{aligned}$$

because $E(t) = 0$ for $t > 0$.

Then:

$D(x)$

$$\begin{aligned} &= \frac{H_D}{\sigma\sqrt{2\pi}} \int_{-\infty}^0 \exp\left[\frac{-(x-t)^2}{2\sigma^2}\right] e^{t/\beta} dt \\ &= \frac{H_D}{\sigma\sqrt{2\pi}} \int_{-\infty}^0 \exp\left[\frac{-\left(x^2 - 2t\left(x + \frac{\sigma^2}{\beta}\right) + t^2\right)}{2\sigma^2}\right] dt \\ &= \frac{H_D}{\sigma\sqrt{2\pi}} \int_{-\infty}^0 \exp\left[\frac{-\left[t^2 - 2\left(x + \frac{\sigma^2}{\beta}\right)t + \left(x + \frac{\sigma^2}{\beta}\right)^2 - \frac{2x\sigma^2}{\beta} - \frac{\sigma^4}{\beta^2}\right]}{2\sigma^2}\right] dt \\ &= \frac{H_D}{\sigma\sqrt{2\pi}} \int_{-\infty}^0 \exp\left[\frac{-\left(x + \frac{\sigma^2}{\beta} - t\right)^2}{2\sigma^2}\right] \exp\left(\frac{x}{\beta}\right) \exp\left(\frac{\sigma^2}{2\beta^2}\right) dt. \end{aligned}$$

We set

$$s = -\left(x + \frac{\sigma^2}{\beta} - t\right)/\sqrt{2}\sigma$$

$$ds = \frac{dt}{\sqrt{2}\sigma}$$

and

$$s^2 = \frac{\left(x + \frac{\sigma^2}{\beta} - t\right)^2}{2\sigma^2}.$$

As

$$t \rightarrow 0, \quad s \rightarrow -\frac{\left(x + \frac{\sigma^2}{\beta}\right)}{\sqrt{2}\sigma} = -Z.$$

Hence:

$D(x)$

$$\begin{aligned} &= \frac{H_D}{\sigma\sqrt{2\pi}} \exp\left(\frac{x}{\beta}\right) \exp\left(\frac{\sigma^2}{2\beta^2}\right) \int_{-\infty}^{-Z} \exp(-s^2) \sqrt{2}\sigma ds \\ &= \frac{H_D}{2} \exp\left(\frac{x}{\beta}\right) \exp\left(\frac{\sigma^2}{2\beta^2}\right) \frac{2}{\sqrt{\pi}} \int_{-\infty}^{-Z} \exp(-s^2) ds. \end{aligned}$$

The symmetry relations for the error function give us

$$\begin{aligned} \frac{2}{\sqrt{\pi}} \int_{-\infty}^{-Z} \exp(-s^2) ds &= \frac{2}{\sqrt{\pi}} \left[\int_0^{\infty} - \int_0^Z \right] \exp(-s^2) ds \\ &= \frac{2}{\sqrt{\pi}} \int_Z^{\infty} \exp(-s^2) ds \\ &= \operatorname{erfc}(Z). \end{aligned}$$

Thus

$$D(x) = \frac{H_D}{2} \exp\left(\frac{x}{\beta}\right) \exp\left(\frac{\sigma^2}{2\beta^2}\right) \operatorname{erfc}\left[\frac{1}{\sqrt{2}} \left(\frac{x}{\sigma} + \frac{\sigma}{\beta}\right)\right].$$

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